1 Describe the oxidation reactions of propan-1-ol when using a suitable oxidising agent.

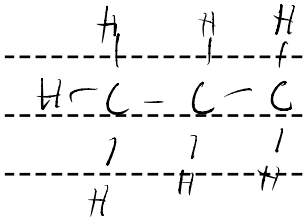
Indicate how the use of different reaction conditions can control which organic product forms.

Include reagents, observations and equations in your answer.

In your equations, use structural formulae and use [O] to represent the oxidising agent.

Propan-1-ol is a primary alcohol. This means it can be oxidised to an aldehyde or carboxylic acid.

An aldehyde can be formed from reacting excess alcohol with an oxidising agent and distilling the aldehyde as soon as it forms.



$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CHO$

Propan-1-ol

If the alcohol is heated under reflux, a carboxylic acid will be produced.

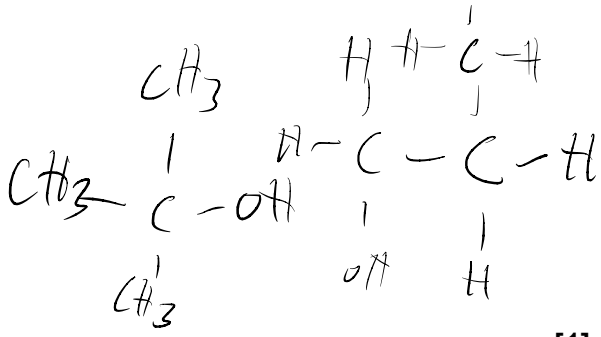
$CH_3CH_2CH_2OH + [O] \rightarrow CH_3CH_2COOH + H_2O$

[6]

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2 Compounds **A** and **B** are structural isomers of $(CH_3)_3COH$.



(i) Compound **A** is a secondary alcohol.

What is the systematic name of compound **A**?

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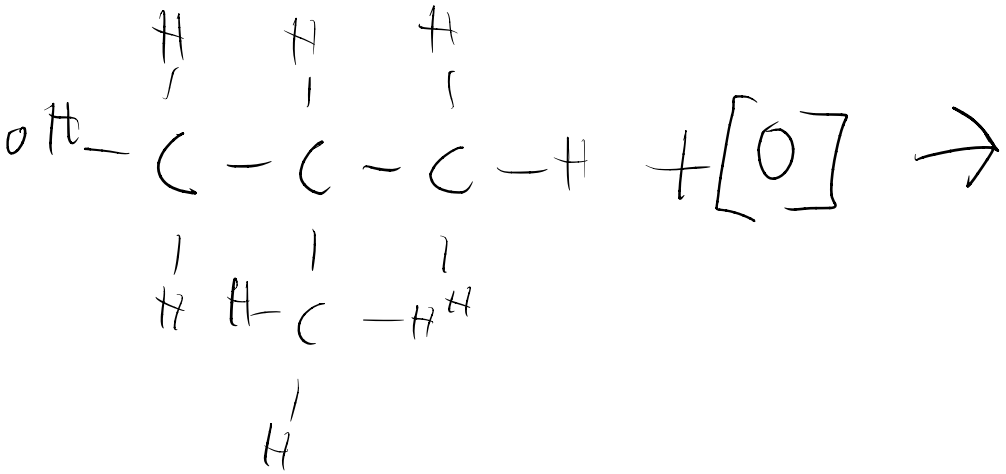
[1]

(ii) Compound **B** is a branched primary alcohol.

Compound **B** is refluxed with acidified potassium dichromate(VI) as an oxidising agent.

Write the equation for the reaction that takes place.

Use structures for organic compounds and [O] for the oxidising agent.



[3]

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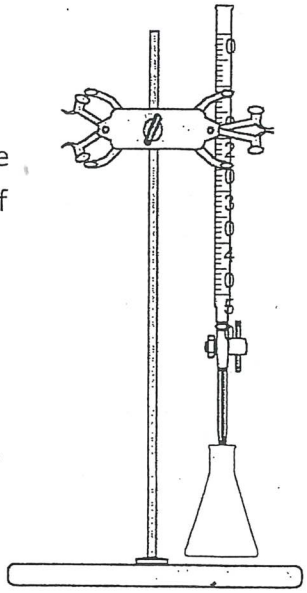
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Fe^{2+}/MnO_4^- Calculations

$$5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

1. A student made up 250cm^3 of an $FeSO_4$ solution in a volumetric flask, using 5g of impure $FeSO_4$. They then titrated 25cm^3 of this with 24.70cm^3 of 0.025 moldm^{-3} $KMnO_4$.
   a. Calculate the number of moles of $KMnO_4$ used
   b. Calculate the number of moles of Fe^{2+} in 25cm^3
   c. Calculate the number of moles of Fe^{2+} in 250cm^3
   d. Calculate the mass of $FeSO_4$ in the sample.
   e. Calculate the % purity of the sample.



~~2~~ A student made up a 250cm^3 solution of $FeSO_4.XH_2O$ using 4.20g of the hydrated salt. He then titrated 25cm^3 of the solution with 19.95cm^3 of 0.015moldm^{-3} $KMnO_4$.

   a. Calculate the number of moles of $KMnO_4$ used
   b. Calculate the number of moles of Fe^{2+} in 25cm^3
   c. Calculate the number of moles of Fe^{2+} in 250cm^3
   d. Work out the Mr of the hydrated salt and hence the value of X.

3. [redacted]

A student made up a 250cm^3 solution of $FeSO_4.XH_2O$ using 3.56 of the **impure** hydrated salt. He then titrated 25cm^3 of the solution with 22.80cm^3 of 0.01moldm^{-3} $KMnO_4$.

   a. Calculate the number of moles of $KMnO_4$ used
   b. Calculate the number of moles of Fe^{2+} in 25cm^3
   c. Calculate the number of moles of Fe^{2+} in 250cm^3
   d. Calculate the mass of Fe^{2+} [redacted] in the sample.
   e. Calculate the % Fe^{2+} in the sample

4. Which sample contains the most iron by mass?

A: 2g of $Fe(OH)_2$

B: 3g of FeO

1. $5Fe^{2+} + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$

2. moles
mass
Conc
vol

3. $FeSO_4 \cdot xH_2O$

	$FeSO_4$	$kmnO_4$
moles		
Conc		
vol		